

Distributed Parallelization of Orthogonalization and Diagonalization in `filter_mpi`

FilterBSE — MPI ring / Gram-matrix implementation

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Abstract

The filter-diagonalization pipeline has three stages: **(1) Filter**, **(2) Orthogonalize**, and **(3) Diagonalize**. Stage 1 is embarrassingly parallel and already MPI-distributed. This note documents the new distributed implementations of stages 2 and 3 (`dist_linalg.c`), which keep the filtered states *column-distributed* across MPI ranks – never gathering the full $N_{\text{grid}} \times N$ state matrix onto a single node – and reduce every large operation to (i) local BLAS-3 products and (ii) communication of only tiny $N \times N$ matrices. No ScaLAPACK is required. The orthogonalization is a distributed *singular value decomposition via the Gram matrix*; the diagonalization distributes the basis transformations (“integrals”) that dominate its cost.

1 Data layout: column (cyclic-block) distribution

Let the matrix of filtered states be

$$A = [a_0 \ a_1 \ \cdots \ a_{N-1}] \in \mathbb{C}^{M \times N}, \quad M = N_{\text{grid}} \sim 10^8, \quad N \sim 10^3,$$

where each *column* $a_i \in \mathbb{C}^M$ is one full-grid wavefunction and $M = \text{nspinngrid}$ is the (huge) number of grid degrees of freedom. The matrix is extremely *tall and skinny*. With P MPI ranks, rank p owns a contiguous block of n_p columns,

$$A_p = A[:, C_p] \in \mathbb{C}^{M \times n_p}, \quad C_p = [o_p, o_p + n_p), \quad \sum_{p=0}^{P-1} n_p = N,$$

so global state g lives on rank $\lfloor g/n \rfloor$ (uniform $n_p = n$ out of the filter). This is exactly the distribution the Filter stage produces, and exactly what the Hamiltonian application needs (each FFT-based $\hat{H}|\psi\rangle$ requires one *whole* state on one rank). It is preserved throughout.

The physical inner product carries the grid volume element $dv = dx dy dz$:

$$\langle x, y \rangle = dv \sum_{g=0}^{M-1} \overline{x_g} y_g, \quad \text{so} \quad S = dv A^H A \in \mathbb{C}^{N \times N}, \quad S_{ij} = \langle a_i, a_j \rangle. \quad (1)$$

2 The two ring primitives

Both stages reduce to two operations on column-distributed data. Each is realised by circulating the large column blocks once around the rank ring (Figure 2), doing a local `zgemm` at every hop, while only tiny $N \times N$ data is ever reduced.

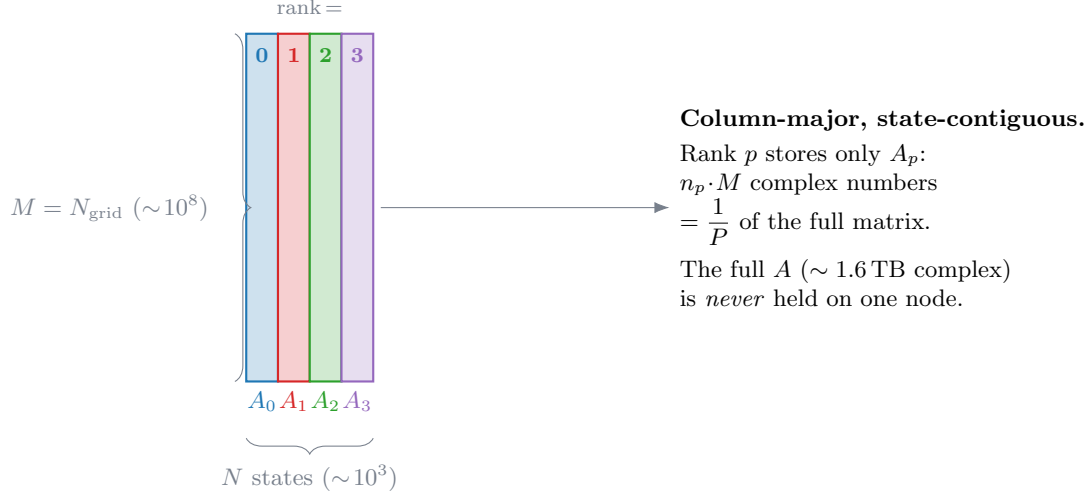


Figure 1: Column-block distribution of the tall-skinny state matrix A . Each rank owns whole columns (states), full grid length each.

(P1) Distributed Gram / overlap matrix. $S = \text{dv } A^H B$ with A, B identically column-distributed. Rank p holds its panel B_p *fixed* and receives every A_q in turn; the local product

$$S[\mathcal{C}_q, \mathcal{C}_p] = \text{dv } A_q^H B_p \in \mathbb{C}^{n_q \times n_p} \quad (\text{one } \mathbf{zgemm}, \text{ contraction over the } M \text{ rows}) \quad (2)$$

fills the column panel $S[:, \mathcal{C}_p]$. After P hops an `MPI.Allgatherv` of the panels assembles the full S on every rank. For orthogonalization $B = A$; for diagonalization $B = \hat{H}U$.

(P2) Distributed back-transform. $\text{out} = A W$ with $W \in \mathbb{C}^{N \times r}$ small and replicated. Rank p accumulates its output columns $\text{out}[:, \mathcal{K}_p]$ by receiving each A_q and adding

$$\text{out}[:, \mathcal{K}_p] += A_q W[\mathcal{C}_q, \mathcal{K}_p] \quad (\text{one accumulating } \mathbf{zgemm} \text{ per hop}). \quad (3)$$

The output is left column-distributed, ready for the next stage.

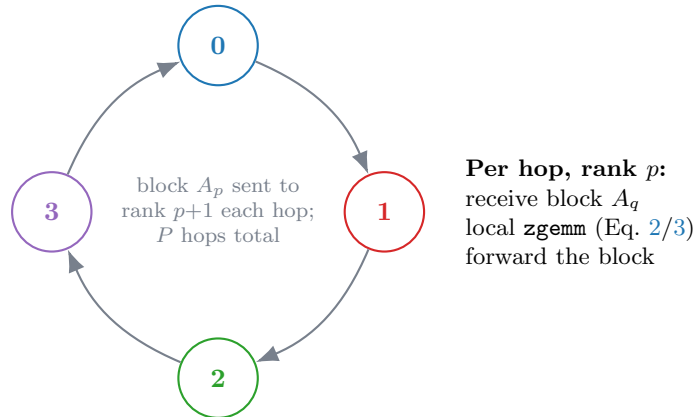


Figure 2: Ring communication. The large column blocks circulate once ($P-1$ neighbour exchanges via `MPI.Sendrecv_replace`); total data moved $= O(MN)$, i.e. the full matrix once, independent of P . Local flops per primitive are $O(MN^2/P)$.

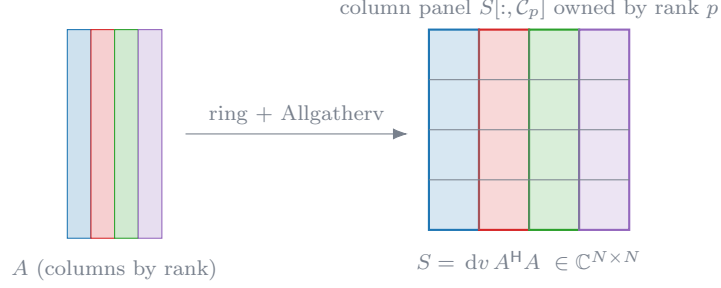


Figure 3: Primitive P1: each rank builds one *column panel* of the small $N \times N$ Gram matrix from local products (Eq. 2); an **Allgatherv** yields the full S (Hermitian) on every rank.

3 Stage 2 — Distributed orthogonalization (SVD via the Gram matrix)

We want an orthonormal basis of the filtered states, discarding numerically redundant directions using the singular-value threshold **SVDEPS**. For an $M \gg N$ matrix, forming the Gram matrix and diagonalizing it is the standard tall-skinny SVD and maps perfectly onto the column layout.

Step 1 — Gram matrix (P1). Equation (1), formed by the ring. Because $A^H A = V \Lambda V^H$ shares its right singular vectors and squared singular values with the SVD $A = U_A \Sigma V^H$, one small Hermitian eigensolve gives the full spectral information:

$$S = V \Lambda V^H, \quad \Lambda = \text{diag}(\lambda_1 \geq \dots \geq \lambda_N \geq 0), \quad \sigma_i = \sqrt{\lambda_i}. \quad (4)$$

This $N \times N$ eigenproblem is solved with LAPACK **zheev** on the root and broadcast (so every rank is bit-identical).

Step 2 — Rank / cutoff. Retain the leading r directions,

$$r = \#\left\{i : \frac{\sigma_i}{\sigma_1} \geq \varepsilon\right\}, \quad \varepsilon = \max(\text{SVDEPS}, \varepsilon_{\text{floor}}), \quad \varepsilon_{\text{floor}} = 10^{-7}. \quad (5)$$

The floor is essential: a Gram (normal-equations) method squares the condition number, so it cannot resolve singular values below $\sqrt{\varepsilon_{\text{mach}}} \approx 1.5 \times 10^{-8}$ relative to σ_1 – below that the squared spectrum $\lambda_i = \sigma_i^2$ sinks under the eigensolver noise floor. Clamping ε to 10^{-7} guarantees that retained directions are genuine, not noise. (A finer cutoff at scale would require a QR-based TSQR instead of the Gram matrix.)

Step 3 — Orthonormal basis (P2). Build the small back-transform matrix and apply it via the ring:

$$W = V_r \Lambda_r^{-1/2} \in \mathbb{C}^{N \times r}, \quad U = A W = A V_r \Lambda_r^{-1/2}, \quad (6)$$

where V_r, Λ_r keep the leading r columns/values. The columns of U are orthonormal in the weighted inner product:

$$\begin{aligned} \langle U, U \rangle &= \text{dv } U^H U = \Lambda_r^{-1/2} V_r^H (\text{dv } A^H A) V_r \Lambda_r^{-1/2} = \Lambda_r^{-1/2} V_r^H S V_r \Lambda_r^{-1/2} \\ &= \Lambda_r^{-1/2} \Lambda_r \Lambda_r^{-1/2} = I_r. \end{aligned} \quad (7)$$

Step 4 — Second pass (polish). To remove the residual non-orthonormality that the squared conditioning leaves in the tail, the whole procedure is repeated once with $A \leftarrow U$ (a CholeskyQR2-style refinement). The second Gram is $\approx I_r$, so it does not truncate further but restores $\langle U, U \rangle = I_r$ to machine precision:

$$U^{(2)} = U^{(1)} V_r^{(2)} (\Lambda_r^{(2)})^{-1/2}, \quad \left\| \text{dv } U^{(2)\text{H}} U^{(2)} - I_r \right\|_{\max} \lesssim 10^{-15}.$$

The output $U \in \mathbb{C}^{M \times r}$ is column-distributed over a fresh block partition $\mathcal{K}_p$ of the r retained states.

4 Stage 3 — Distributed diagonalization

The eigenvectors are computed in the small orthonormal basis U and transformed back to the grid. Every large operation is again local + one small reduction.

Step 1 — Apply the Hamiltonian (local). Each rank applies $\hat{H}$ (kinetic via FFT, local potential, non-local/spin-orbit projectors) to *its own* states – embarrassingly parallel, reusing `p_hamiltonian`:

$$(\hat{H}U)_p = \hat{H}U_p, \quad k \in \mathcal{K}_p.$$

Step 2 — Reduced Hamiltonian (P1). Form the $r \times r$ matrix with the Gram primitive using $B = \hat{H}U$:

$$\tilde{H} = \text{dv } U^{\text{H}} \hat{H} U \in \mathbb{C}^{r \times r}, \quad \tilde{H}_{ij} = \langle u_i, \hat{H} u_j \rangle, \quad \tilde{H} = \tilde{H}^{\text{H}}. \quad (8)$$

Step 3 — Small eigenproblem. On the root, then broadcast:

$$\tilde{H} = C E C^{\text{H}}, \quad E = \text{diag}(E_1 \leq \dots \leq E_r), \quad (9)$$

so E are the quasiparticle energies and $C \in \mathbb{C}^{r \times r}$ the eigenvectors in the filtered basis.

Step 4 — Back to the grid basis (P2).

$$\Psi = U C, \quad \psi_k = \sum_{i=1}^r u_i C_{ik}, \quad (10)$$

again a ring back-transform, leaving Ψ column-distributed.

Step 5 — Eigenvalue variance (local). The ghost-state diagnostic is purely local because each rank owns whole states. Using Hermiticity, $\langle \psi | \hat{H}^2 | \psi \rangle = \|\hat{H}\psi\|^2$, so a *single* extra Hamiltonian application suffices:

$$\sigma_{E,k}^2 = \langle \psi_k | \hat{H}^2 | \psi_k \rangle - \langle \psi_k | \hat{H} | \psi_k \rangle^2 = \|\hat{H}\psi_k\|^2 - E_k^2. \quad (11)$$

Output. Eigenvectors are written to `psi.dat` with collective MPI-IO: state g is placed at byte offset $g \cdot \text{stlen} \cdot 8$ with `stlen = complex_idx \cdot M`, matching the serial writer and the `get_n.states.x`/BSE readers. Each rank writes its block $\mathcal{K}_p$ directly at its global offset; energies and σ_E are gathered to the root for `eval.dat`.

5 Cost and communication

Operation	Local flops	Communication	Reduced size
Gram $S = \text{dv } A^H A$ (P1)	$O(MN^2/P)$	$O(MN)$ ring + Allgatherv	$N \times N$
Back-transform $A W$ (P2)	$O(MNr/P)$	$O(MN)$ ring	—
Small eigensolve (zheev)	$O(N^3)$ (root)	Bcast $N \times N$	$N \times N$
$\hat{H}U$ (per state)	$O(\frac{N}{P} M \log M)$	none	—

Because $N \sim 10^3$, the $O(MN^2)$ compute dominates the $O(MN)$ communication by a factor $\sim N$: the ring exchanges are cheap relative to the GEMMs, and no operation ever materialises more than $O(MN/P)$ (one rank’s share) of the large data. Memory per rank stays at a small multiple of A_p .

6 Validation

Test	Metric	Result
Unit: ring Gram vs. dense (4 ranks)	$\max S_{\text{ring}} - S_{\text{ref}} $	1.6×10^{-13}
Unit: distributed SVD, rank-deficient	retained rank r	correct (uneven split)
Unit: orthonormality	$\max \text{dv } U^H U - I $	2.4×10^{-15}
End-to-end CsPbI ₃ (4 ranks) vs. serial	converged eigenvalues	match to 1.6×10^{-6} eV
End-to-end: psi.dat	byte layout / offsets	exact

The distributed path retains slightly fewer states than the serial **SVDEPS** = 10^{-10} SVD (the tail trimmed by the 10^{-7} Gram floor of Eq. 5); those are the ill-conditioned directions the σ_E ghost filter removes anyway, and all converged physical eigenvalues agree with the serial reference to $\sim 10^{-6}$ eV.